Data Augmentation for Image Classification

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I. INTRODUCTION

In this work, we study the effect of data augmentation on scene classification using the Scenes15 dataset. The goal is to improve the generalization performance of a SimpleCNN model.

First, the model was trained without augmentation to obtain a baseline. The results showed strong overfitting, with 99.44% training accuracy and only 65.89% test accuracy.

To reduce overfitting, several augmentation techniques were tested using the Albumentations library, including HorizontalFlip, RandomResizedCrop, Rotate, and ColorJitter. Among all tested methods, HorizontalFlip provided the biggest improvement, increasing test accuracy to 70%. The combination of HorizontalFlip and RandomResizedCrop proved to be just as efficient. Finally, MixUp was implemented as an additional regularization technique to further improve generalization.

II. METHOD

The experiments were performed using a SimpleCNN architecture trained on the Scenes15 dataset. The dataset contains indoor and outdoor scenes such as kitchens, offices, forests, streets and natural environments.

Data augmentation was implemented using the Albumentations library. The following transformations were evaluated:

HorizontalFlip: This transformation is particularly suitable for Scenes15, since mirroring a scene does not change its semantic meaning. Flipping an image of a room gives us a different, mirrored room.

RandomResizedCrop: Scene images can be taken from slightly different distances or framings, so cropping the images seemed like a good idea to experiment.

Rotate: A small rotation range (±5 degrees) was applied to simulate slight camera misalignment in real-world photographs.

ColorJitter: Since the dataset is effectively grayscale, only brightness and contrast variations are meaningful. This transformation was used to simulate different illumination conditions.

InvertedImage: This transformation inverts pixel intensities, producing a negative version of the grayscale image. It was included to evaluate whether the model relies mainly on edges and shapes rather than intensity values.

Each transformation was first tested individually. Later, combinations of the most promising transformations were evaluated to analyze their joint effect on performance.

III. IMPLEMENTATION

A. Data Augmentation Pipeline

Data augmentation was implemented using the Albumentations library. A function named `get_training_augmentation()` was created to define the selected transformations. The function takes no input and returns an Albumentations Compose object applied to each training image.

B. Training Procedure

The SimpleCNN architecture provided in the tutorial was used. The network was trained for 20 epochs using the Adam optimizer.

C. MixUp Implementation

For Question 4, a function named `mixup_data(x, y, alpha)` was implemented. This function mixes two images from the same batch using a coefficient λ sampled from a Beta distribution.

D. Running the code

The experiments were implemented in a Jupyter Notebook environment using Kaggle.

To run the code, it is sufficient to open the notebook and execute all cells sequentially. The required libraries (PyTorch, Torchvision, and Albumentations) must be installed before running the notebook.

IV. DATA

The experiments were conducted on the Scenes15 dataset, which contains 4,485 RGB images divided into 15 scene categories. The dataset includes both indoor and outdoor environments, such as kitchens, offices, forests, streets and coastal scenes. The dataset was randomly split into 80% training data and 20% test data. All images were resized to 128×128 pixels before being used as input to the network.

V. RESULTS AND ANALYSIS

A. Baseline (No Augmentation)

Without any data augmentation, the model achieved:

- Train Accuracy: 99.44%
- Test Accuracy: 65.89%

These results show strong overfitting, since the model performs very well on the training data but poorly on unseen test data.

B. Individual Transformations

HorizontalFlip (p=0.5)

- Train Accuracy: 88.85%
- Test Accuracy: 69.45%

This transformation provided the largest improvement. Test accuracy increased and overfitting decreased. Since flipping a scene does not change its semantic meaning, this augmentation increases data diversity without altering structure.

RandomResizedCrop

- Train Accuracy: 91.33%
- Test Accuracy: 69.79%

Cropping may remove important visual elements from the scene, limiting the effectiveness, but this transformation showed better test accuracy than the baseline.

Rotate (limit=5)

- Train Accuracy: 92.25%
- Test Accuracy: 62.54%

Small rotations slightly reduced overfitting, but test accuracy decreased compared to the baseline. This suggests that small geometric misalignment is not a significant challenge in this dataset.

ColorJitter

- Train Accuracy: 98.22%
- Test Accuracy: 65.22%

Brightness and contrast variations had limited impact. Although scenes can appear under different lighting conditions, the results suggest that spatial structure is more important than intensity variations for scene classification.

InvertedImage

- Train Accuracy: 94.37%
- Test Accuracy: 68.90%

This transformation inverts pixel intensities, producing a negative version of the grayscale image. The results show an improvement compared to the baseline, suggesting that the model relies more on structural information than on absolute intensity values. However, overfitting remains relatively high, and performance is slightly lower than the best geometric transformations.

C. Combination of Transformations

Several combinations of the most promising transformations were tested.

HorizontalFlip + RandomResizedCrop

- Train Accuracy: 84.62%
- Test Accuracy: 70.46%

This combination produced the best overall performance. The two transformations complement each other: HorizontalFlip increases viewpoint invariance, while RandomResizedCrop introduces framing variability. This indicates a positive synergy between geometric transformations, leading to improved generalization and reduced overfitting.

HorizontalFlip + RandomResizedCrop + Rotate

- Train Accuracy: 83.25%
- Test Accuracy: 69.45%

We observed that the order of the augmentations is important. If rotation is applied before cropping, padding may appear in the image. In our case, the transformations were applied in an order that avoids such problems. Adding rotation slightly decreased test accuracy compared to HorizontalFlip + RandomResizedCrop.

HorizontalFlip + RandomResizedCrop + InvertedImage

- Train Accuracy: 75.56%
- Test Accuracy: 67.67%

This combination reduced test accuracy compared to the best two-transform setup. However, the gap between training and test accuracy became smaller, indicating a strong reduction in overfitting. Although InvertedImage alone improved performance over the baseline, combining it with other strong augmentations did not lead to the highest accuracy.

InvertedImage + HorizontalFlip + RandomResizedCrop + Rotate

- Train Accuracy: 75.81%
- Test Accuracy: 65.44%

When all transformations were combined, performance decreased further. Although overfitting remained relatively low, the test accuracy dropped compared to simpler combinations. Applying many augmentations together may introduce conflicting effects. For example, rotation can create padding artifacts, and intensity inversion may amplify these artificial regions. Even though the order of transformations was chosen carefully to minimize such issues, excessive augmentation appears to reduce performance.

D. MixUp

As an additional experiment, the MixUp technique was implemented during training.

With MixUp, the model reached a train accuracy of 87.79% and a test accuracy of 69.57%. This shows an improvement compared to the baseline model without augmentation, but the performance was still slightly lower than the best geometric augmentation combination (HorizontalFlip + RandomResizedCrop).

These results suggest that MixUp helps the model generalize better than the baseline, but geometric transformations remain more effective for this scene classification task.

VI. CONCLUSIONS

Thus, among the individual transformations, HorizontalFlip and RandomResizedCrop produced the best improvements. Their combination achieved the highest test accuracy and showed a clear positive synergy between geometric augmentations.

More complex combinations with many transformations reduced performance, suggesting that excessive augmentation may distort the data distribution. MixUp also improved generalization compared to the baseline, but it did not outperform the best geometric augmentation setup.

Overall, the results show that moderate geometric augmentations are the most effective strategy for improving scene classification performance in this dataset.

VII. TIME LOG

We tried different strategies, each student on his own, but also together. In the end we discussed and compared our results, choosing the best methods and ideas. Here is the time we spent on working:

- Andrea
  - Study: 1h
  - Coding: 1h
  - Writing: 1h
  - Testing: 3h
- Alexandra
  - Study: 1h
  - Coding: 2h
  - Writing: 3h
  - Testing: 1h
- Adham
  - Study: 2h
  - Coding: 1h
  - Writing: 1h
  - Testing: 2h